

Networking Fundamentals

From what an IP address is, to designing a real Azure virtual network

Module	0 - the foundation for everything that follows
Starts from	Zero. No networking background assumed.
Leads into	Module 5 (Virtual Networks) and Module 6 (Inter-site Connectivity)
Contains	26 worked examples and 20 practice problems with full solutions
How to use it	Read Part A and B before class. Work Part C with a pen. Keep the last page next to you during every lab.

Contents

Part A	Foundations	What a network is, how a message travels, the devices involved
Part B	IP addresses	Anatomy, binary, public and private, static and dynamic
Part C	Subnetting	Masks, CIDR, the three-step method, splitting a network
Part D	Azure networking	Virtual networks, subnets, the five reserved addresses, NSGs
Part E	Practice	Drills, problems with solutions, glossary, one-page cheat sheet

Read this before the Virtual Network lab

You can build a virtual network in Azure without understanding any of this. It will work. Then you will add a second subnet, get an overlap error, and have no idea why. This module is what prevents that.

PART A

1. Why This Module Comes First

When you create one virtual machine in Azure, the platform quietly creates a virtual network, a subnet, a network interface, a public IP address and a network security group alongside it. Five of those six things are networking. If you do not understand them, you do not really understand what you just built.

Here is the practical argument. Every symptom in the left column looks like a different problem to a beginner. Every one of them is the same problem.

What the student says	What is actually wrong
"My SSH just hangs and times out"	A security rule is not allowing port 22
"The two networks refuse to connect"	Their address ranges overlap
"It will not let me add another subnet"	The address space is already used up
"I made a /29 but only 3 VMs fit"	Azure reserves five addresses in every subnet
"My website is not opening from home"	No public IP, or port 80/443 is not open
"These two VMs cannot ping each other"	They are in different networks, or a rule blocks it
"The VM has no internet access"	No outbound path configured

Ek line mein

Azure me jitni bhi "kaam nahi kar raha" wali problem aati hai, usme se zyadatar networking ki hoti hai. Isliye ye module sabse pehle aa raha hai.

2. What a Network Actually Is

A network is two or more devices that can send data to each other. That is genuinely the whole idea. Everything after this is machinery to make it work reliably at large scale.

The postal analogy

Think about sending a letter. Three pieces of information are needed and they map exactly onto networking.

Sending a letter	In networking	Example
The house address	The IP address - which machine	10.0.1.7
The person's name on the envelope	The port number - which application on that machine	22
The post office you drop it at	The default gateway - your way out of the local network	10.0.1.1
The postal network in between	Routers - passing it along until it arrives	many hops

Ek line mein

IP address batata hai **kaunsi machine**, port batata hai **us machine pe kaunsa program**, aur gateway batata hai **bahar nikalne ka rasta**.

3. How a Message Actually Travels

Follow one request end to end. Your laptop wants to open a website. Here is what happens, in order, with nothing skipped.



Same street? Send directly. Different street? Hand it to the gateway.

The path a request takes. The important decision happens at the very first step.

- 1 Your laptop asks one question first: is the destination on my own street?** It compares the destination address against its own address and subnet mask. This single comparison decides everything that follows.
- 2 If the answer is yes** - the destination is in the same subnet - the laptop sends the data directly. No router is involved at all. This is why two virtual machines in the same Azure subnet can talk without you configuring anything.
- 3 If the answer is no**, the laptop cannot reach it alone. It hands the data to its default gateway, which is the router. In Azure, that gateway is always the second address of your subnet - 10.0.1.1 in a 10.0.1.0/24 subnet.
- 4 The router looks at its routing table** and decides where to forward the data next. It does not know the whole path, only the next step. Each router along the way repeats this.
- 5 The data crosses the internet** through however many routers it takes, each making the same next-step decision.
- 6 It arrives at the destination machine**, which reads the port number to decide which application should receive it. Port 443 goes to the web server, port 22 to the SSH server.
- 7 The reply travels back** the same way, making the same decisions in reverse.

The one idea to take from this section

Every machine makes exactly one routing decision: same subnet, or not? If yes, deliver directly. If no, hand it to the gateway. Everything about subnets and masks exists to let a machine answer that question.

Ek line mein

Har machine sirf ek sawaal poochti hai - destination meri hi gali me hai ya nahi? Haan to seedha bhej do, nahi to gateway ko de do.

4. The Devices, and Their Azure Equivalents

In a physical office you would see these boxes in a rack. In Azure you never see them, but the same functions are there as software you configure.

Physical device	What it does	In Azure
Switch	Connects machines on the same local network. Passes data by hardware address.	Invisible. The platform handles it inside a subnet.
Router	Connects different networks. Decides the next hop.	Built into the virtual network. Custom routes are Route Tables.
Firewall	Allows or blocks traffic based on rules.	Network Security Group for basic rules, Azure Firewall for advanced.
Load balancer	Spreads incoming traffic across several servers.	Azure Load Balancer at Layer 4, Application Gateway at Layer 7.
DNS server	Turns names into IP addresses.	Azure DNS, or the platform resolver at 168.63.129.16.
NAT device	Lets private machines reach the internet using a shared public address.	NAT Gateway, or default outbound access.

5. Layers - The Short Honest Version

Textbooks present seven layers. You will be examined on the names, but only three of them affect your daily work. Learn the working version first and the names will stick.

The idea is simple: as data leaves a machine it gets wrapped in envelopes, one inside another. As it arrives it gets unwrapped in reverse. Each layer only cares about its own envelope.

Layer	Name	Deals with	What you configure in Azure
7	Application	HTTP, DNS, SSH - the actual service	Application Gateway, Azure DNS
6	Presentation	Formatting and encryption	Nothing directly
5	Session	Keeping a conversation open	Nothing directly
4	Transport	Ports, TCP and UDP, reliable delivery	NSG rules, Load Balancer rules
3	Network	IP addresses, routing between networks	Virtual networks, subnets, route tables
2	Data link	Hardware addresses on the local segment	Nothing. Azure handles it.
1	Physical	Cables, signals, radio	Nothing. That is Microsoft's data centre.

A shortcut worth memorising for the exam

Azure Load Balancer works at Layer 4 - it decides using IP addresses and ports only. Application Gateway works at Layer 7 - it can read the URL and route on that. Whenever a question asks which one to use, it is really asking whether the decision needs to read the URL.

Ek line mein

Tumhe rozana sirf teen layer chhuni hain - Layer 3 (IP aur routing), Layer 4 (ports aur NSG rules), aur Layer 7 (URL-based routing). Baaki Azure khud sambhaal leta hai.

PART B

6. The IP Address, Piece by Piece

An IPv4 address is a 32-bit number. Humans are bad at reading 32 binary digits, so it is written as four decimal numbers separated by dots. Each part is called an octet because it holds 8 bits. Eight bits can represent 0 to 255, which is why no octet ever goes above 255.



10.0.1.7 with a /24 prefix. The first three octets say which network; the last says which machine inside it.

Why 255 is the ceiling

Eight bits, all set to 1, gives $128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 255$. There is no way to write 256 in eight bits. If you ever type 10.0.1.300 into Azure it will be rejected, and now you know exactly why.

Ek line mein

IP address char hisso me likha jaata hai, har hissa 0 se 255 tak. 255 se upar isliye nahi ja sakte kyunki 8 bits me usse zyada aata hi nahi.

7. Binary - Learn One Octet and You Know Them All

You will not do binary arithmetic in the Azure portal. But subnetting only makes sense in binary, because a subnet boundary is a position inside that 32-bit string. Twenty minutes here saves hours later.

The eight positions

Every octet has eight positions. Each position has a fixed value. From left to right they halve each time.

Position value:	128	64	32	16	8	4	2	1	
Bit:	1	1	1	1	1	1	1	1	= 255

Decimal to binary - the subtraction method

Work left to right. At each position ask: does this value fit into what is left? If yes, write 1 and subtract it. If no, write 0 and move on.

Worked example: convert 200

Worked example 1

Start with 200.

128 fits into 200?	Yes	-> write 1,	200 - 128 = 72
64 fits into 72?	Yes	-> write 1,	72 - 64 = 8
32 fits into 8?	No	-> write 0	
16 fits into 8?	No	-> write 0	
8 fits into 8?	Yes	-> write 1,	8 - 8 = 0
4 fits into 0?	No	-> write 0	
2 fits into 0?	No	-> write 0	
1 fits into 0?	No	-> write 0	

Answer: 200 = 11001000

Worked example: convert 10

Worked example 2

128? No -> 0	64? No -> 0	32? No -> 0	16? No -> 0
8? Yes -> 1, remainder 2			
4? No -> 0			
2? Yes -> 1, remainder 0			
1? No -> 0			

Answer: 10 = 00001010

Binary to decimal - just add

Add up the position values wherever there is a 1.

Worked example 3

```
11000000 -> 128 + 64 = 192
11111000 -> 128 + 64 + 32 + 16 + 8 = 248
10101100 -> 128 + 32 + 8 + 4 = 172
```

The values you will meet over and over

Subnet masks only ever use these nine values in an octet. Memorising this short list is enough for almost every question you will face.

Decimal	Binary	Bits set	Appears in
0	00000000	0	/8, /16, /24 masks
128	10000000	1	/25
192	11000000	2	/26
224	11100000	3	/27
240	11110000	4	/28
248	11111000	5	/29
252	11111100	6	/30
254	11111110	7	/31
255	11111111	8	/8, /16, /24 masks

Ek line mein

Sirf ye nau numbers yaad kar lo - 0, 128, 192, 224, 240, 248, 252, 254, 255. Subnet mask me inke alawa kuch aata hi nahi.

8. Public and Private Addresses

Not every address is reachable from the internet. Three ranges are reserved for private use. Anyone may use them inside their own network, and no router on the public internet will carry them. These are the ranges you will use for every Azure virtual network you ever build.

Range	In CIDR	How many	Typical use
10.0.0.0 to 10.255.255.255	10.0.0.0/8	16,777,216	Large networks. The usual choice in Azure.
172.16.0.0 to 172.31.255.255	172.16.0.0/12	1,048,576	Medium networks. A common corporate default.
192.168.0.0 to 192.168.255.255	192.168.0.0/16	65,536	Home and small office routers.

How to tell at a glance

- 1 Does it start with **10.** ? Private.
- 2 Does it start with **192.168.** ? Private.
- 3 Does it start with **172.** and the second number is between 16 and 31? Private. Note that 172.15 and 172.32 are public - only 16 through 31 are reserved.
- 4 Anything else? Public.

If private addresses are not routable, how does a VM reach the internet?

Through NAT - network address translation. Your machine sends the request with its private address. On the way out, a device swaps the source address for a public one and remembers the swap. The reply comes back to the public address and is translated back to your private one. This is exactly what your home router does for every phone and laptop in your house.

The mistake that costs people a weekend

If your Azure virtual network uses 192.168.1.0/24 and your office router also uses 192.168.1.0/24, you will never connect the two with a VPN. The ranges overlap and routing becomes ambiguous. Before choosing an address space, always ask what the other side already uses. Because almost every home router in India uses 192.168.1.0 or 192.168.0.0, avoid those two ranges in anything that may later connect to a home or branch network.

Ek line mein

Private address ghar ke andar ke room numbers jaise hain - har ghar me Room 1 ho sakta hai. Public address poore sheher me unique hota hai.

9. Static, Dynamic, and the Two Addresses Every VM Has

	Dynamic	Static
Who assigns it	Azure picks the next free address	You choose it and it is reserved
Survives a stop and start?	May change	Stays the same
Use it for	Ordinary servers, test machines	Domain controllers, DNS servers, anything other systems point at by address

	Private IP	Public IP
Where it comes from	The subnet you placed the VM in	A separate Azure resource you attach
Reachable from	Other resources inside the virtual network	The internet
Costs money?	No	Yes, public IP addresses are billed
Needed for a database server?	Yes	Usually no, and adding one is a security mistake
Needed for a public website?	Yes	Yes, or a load balancer in front of it

PART C

10. The Subnet Mask - What It Is Actually For

An IP address on its own does not tell you where the network ends and the machine begins. 10.0.1.7 could be machine 7 on network 10.0.1, or machine 1.7 on network 10.0. The subnet mask settles it.

The mask is another 32-bit number. Every position that is a 1 means "this bit belongs to the network". Every 0 means "this bit identifies the machine". The ones always come first and never mix with the zeros.

The same information written two ways

```
IP address      10.0.1.7      00001010.00000000.00000001.00000111
Subnet mask     255.255.255.0    11111111.11111111.11111111.00000000
                                     |----- network -----| - host -|

Counting the ones: 24.    So we write it as 10.0.1.0/24
```

That count is all CIDR notation is. The /24 replaces writing 255.255.255.0 every time. Azure uses CIDR everywhere, so from here on we will too.

Ek line mein

Mask ka kaam sirf itna hai - ye batana ki address ka kitna hissa **network** hai aur kitna **machine**. /24 ka matlab pehle 24 bits network ke hain.

11. The CIDR Table

This is the single most useful page in the handout. The column you will use most in Azure is the fourth one, because Azure takes five addresses out of every subnet - explained in section 14.

CIDR	Subnet mask	Block size	Total	Usable in Azure	Sensible use
/24	255.255.255.0	256	256	251	A standard subnet. Start here.
/25	255.255.255.128	128	128	123	Half a standard subnet.
/26	255.255.255.192	64	64	59	A small application tier.
/27	255.255.255.224	32	32	27	A database tier of a few servers.
/28	255.255.255.240	16	16	11	Very small. Plan carefully.
/29	255.255.255.248	8	8	3	The smallest Azure allows.
/23	255.255.254.0	512	512	507	Two standard subnets worth.
/22	255.255.252.0	1,024	1,024	1,019	A large tier.
/16	255.255.0.0	-	65,536	-	A whole virtual network address space.

Working it out without the table

The host portion is 32 minus the prefix. Two to that power is the total number of addresses. That is the entire calculation.

```
/24 -> 32 - 24 = 8 host bits    -> 2^8 = 256  
/26 -> 32 - 26 = 6 host bits    -> 2^6 = 64  
/27 -> 32 - 27 = 5 host bits    -> 2^5 = 32  
/29 -> 32 - 29 = 3 host bits    -> 2^3 = 8
```

Powers of two worth knowing: 2 4 8 16 32 64 128 256 512 1024

Bigger number, smaller network

This confuses everyone at first. A /16 is enormous and a /29 is tiny. The number counts how many bits are locked to the network, so a larger number leaves fewer bits free for machines. Every time the prefix goes up by one, the subnet halves.

12. Reading a Subnet - The Three-Step Method

Given any address with a prefix, you can find everything about its subnet with three steps. Learn this method and subnetting stops being frightening.

- 1 Find the block size.** Block size is 256 minus the interesting octet of the mask. Or look it up in the table on the previous page.
- 2 Count in block-size jumps** from zero until you pass the address you were given. The last value you landed on before passing it is the network address.
- 3 The next block start, minus one, is the broadcast address.** Everything between them is the usable range.

Worked example A - what subnet is 10.0.1.100/26 in?

Worked example A

Step 1 /26 mask is 255.255.255.192. Block size = $256 - 192 = 64$

Step 2 Count in 64s: 0, 64, 128, 192
Where does 100 fall? Between 64 and 128.
Network address = 10.0.1.64

Step 3 Next block starts at 128, so broadcast = 10.0.1.127

Result Network 10.0.1.64
Broadcast 10.0.1.127
Range 10.0.1.64 to 10.0.1.127 (64 addresses)
Usable in Azure: 59

Worked example B - what subnet is 192.168.10.200/28 in?

Worked example B

Step 1 /28 mask is 255.255.255.240. Block size = $256 - 240 = 16$

Step 2 Count in 16s: 0, 16, 32, ... 176, 192, 208
200 falls between 192 and 208.
Network address = 192.168.10.192

Step 3 Next block starts at 208, so broadcast = 192.168.10.207

Result Range 192.168.10.192 to 192.168.10.207 (16 addresses)
Usable in Azure: 11

Worked example C - are these two addresses in the same subnet?

Worked example C

Question: 10.0.1.50/26 and 10.0.1.90/26 - same subnet or not?

Block size for /26 = 64. Blocks: 0-63, 64-127, 128-191, 192-255

10.0.1.50 falls in 0-63 -> network 10.0.1.0

10.0.1.90 falls in 64-127 -> network 10.0.1.64

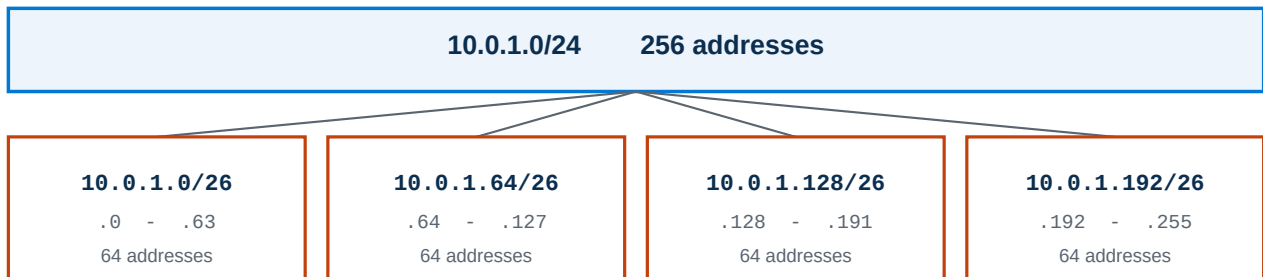
Answer: different subnets. They cannot talk directly - traffic must go through the gateway, and a rule could block it.

Ek line mein

Teen step yaad rakho - block size nikalo, block size ke jumps me gino, aur agla block shuru hone se ek kam hi broadcast hai. Bas itna hi subnetting hai.

13. Splitting a Network Into Subnets

This is the same method used in reverse, and it is what you will actually do when planning an Azure network. You start with one range and cut it into pieces.



Each step of 64 starts a new block. The jump size is the block size.

One /24 cut into four /26 blocks. The jump size and the block size are the same number - 64.

Worked example D - split 10.0.1.0/24 into four equal subnets

Worked example D

Four pieces needed. $4 = 2^2$, so borrow 2 bits. $24 + 2 = /26$
Block size for /26 = 64.

Subnet 1	10.0.1.0/26	range	.0	to	.63
Subnet 2	10.0.1.64/26	range	.64	to	.127
Subnet 3	10.0.1.128/26	range	.128	to	.191
Subnet 4	10.0.1.192/26	range	.192	to	.255

How many bits to borrow

Pieces you want	Bits to borrow	A /24 becomes
2	1	/25
4	2	/26
8	3	/27
16	4	/28
32	5	/29

Subnets do not have to be equal

In real designs they usually are not. A web tier may need a /24 while a database tier needs only a /27. Mixing sizes is normal and is called variable length subnet masking. The only rules are that pieces must not overlap and each must start on a valid boundary for its size.

From 10.0.0.0/16, unequal pieces:

10.0.1.0/24	web tier	256 addresses
10.0.2.0/24	app tier	256 addresses
10.0.3.0/27	database tier	32 addresses
10.0.3.32/27	management	32 addresses

Still free: 10.0.3.64 onwards, and all of 10.0.4.0 upward

PART D

14. The Azure Rule Nobody Warns You About

In every Azure subnet, five addresses are reserved by the platform and cannot be given to your resources. This appears on the exam and it catches people in real projects.

In a 10.0.1.0/24 subnet	Reserved for	Why
10.0.1.0	Network identifier	Standard networking - identifies the subnet itself
10.0.1.1	Default gateway	The way out of the subnet
10.0.1.2	Azure platform DNS	Mapped to the platform resolver
10.0.1.3	Azure platform DNS	Mapped to the platform resolver
10.0.1.255	Broadcast address	Standard networking - the last address in the range

So a /24 offers 256 addresses on paper and 251 in practice. On small subnets the effect is severe: a /29 has 8 addresses, five are reserved, and three machines fit. This is exactly why /29 is the smallest subnet Azure supports - anything smaller would leave nothing at all.

The sizing rule

- 1 Take the number of machines the subnet must hold.
- 2 **Add five.** This is the step everyone forgets.
- 3 Find the smallest prefix whose total is equal to or greater than that number.

Two sizing examples

Requirement: 12 virtual machines

```
12 + 5 = 17 addresses needed
/28 gives 16  -> not enough
/27 gives 32  -> fits.   Answer: /27
```

Requirement: 30 virtual machines

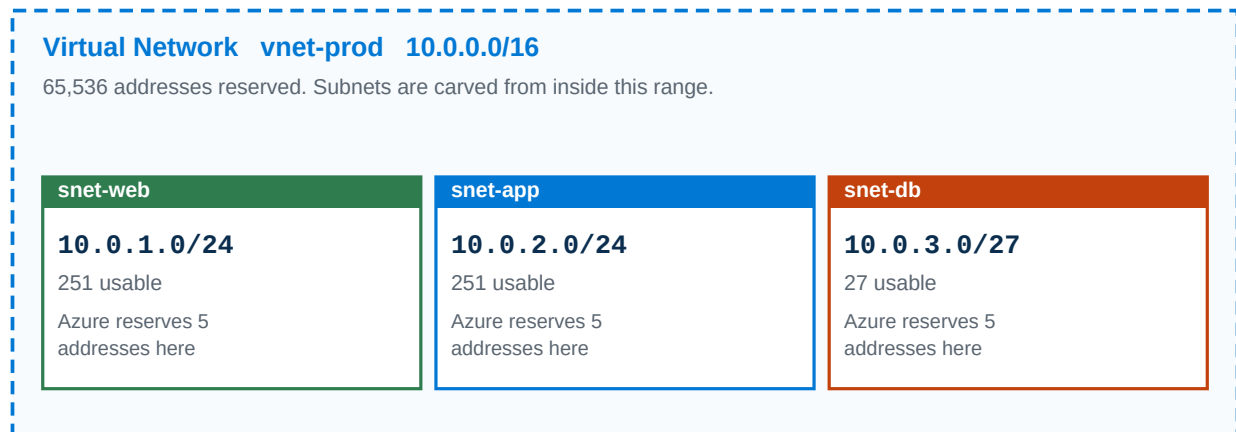
```
30 + 5 = 35 addresses needed
/27 gives 32  -> not enough
/26 gives 64  -> fits.   Answer: /26
```

Ek line mein

Azure har subnet me 5 address khud rakh leta hai. Isliye jitni VMs chahiye, usme 5 jod ke phir prefix chuno. Ye exam me baar-baar poocha jaata hai.

15. Virtual Networks and Subnets in Azure

A virtual network is your private, isolated network inside Azure. You give it an address space. You then carve subnets out of that space, and every resource lives in a subnet.



One virtual network with three subnets. They do not overlap and all sit inside the parent range.

Rule	What it means in practice
A subnet must sit inside the virtual network address space	You cannot put a 10.1.0.0/24 subnet inside a 10.0.0.0/16 network. The portal will reject it.
Subnets in the same network must not overlap	10.0.1.0/24 and 10.0.1.0/25 cannot both exist. Azure rejects overlapping ranges.
Networks you intend to connect must not overlap	Peering and VPN both fail if the ranges collide. Plan this before you build anything.
Resizing a subnet needs it to be empty or nearly so	Generous planning at the start is far cheaper than rebuilding later.
Traffic between subnets in the same network is allowed by default	If you want the database tier isolated from the web tier, you must create rules for it.
Each subnet gets its gateway at the second address	In 10.0.1.0/24 the gateway is always 10.0.1.1. You never configure this.

16. A Complete Worked Design

You are asked to design the network for a three-tier application: a public web tier, an application tier and a database tier. Expected sizes are about 40 web servers, 20 application servers and 4 database servers, with room to grow.

Step 1 - choose the virtual network range

Pick from a private range, large enough for everything with room spare, and not colliding with anything the organisation already uses. A /16 is a sensible default.

Virtual network: vnet-prod 10.20.0.0/16 (65,536 addresses)

Step 2 - size each subnet, remembering the five reserved

Tier	Machines	Plus 5	Smallest that fits	Chosen	Why
Web	40	45	/26 gives 64	/24	Room to grow; address space is free
Application	20	25	/27 gives 32	/24	Consistent sizing is easier to manage
Database	4	9	/28 gives 16	/27	Small tier, but leave some headroom

Step 3 - allocate non-overlapping ranges

```
snet-web    10.20.1.0/24    10.20.1.0   to 10.20.1.255    251 usable
snet-app    10.20.2.0/24    10.20.2.0   to 10.20.2.255    251 usable
snet-db     10.20.3.0/27    10.20.3.0   to 10.20.3.31     27 usable
```

Free for later: 10.20.3.32 onwards, and all of 10.20.4.0 upward

Why leave gaps

Using a /24 for a tier that only needs a /26 looks wasteful, but private address space costs nothing and resizing a live subnet is painful. Leave room. The only real constraint is not colliding with networks you may connect to later.

17. Building the Design in the CLI

The design above, created in three commands. Read them next to the plan and you will see the addresses map straight across.

Azure CLI

```
# 1. The virtual network, with the web subnet created at the same time
az network vnet create \
  --resource-group rg-network \
  --name vnet-prod \
  --address-prefix 10.20.0.0/16 \
  --subnet-name snet-web \
  --subnet-prefix 10.20.1.0/24

# 2. Add the application subnet
az network vnet subnet create \
  --resource-group rg-network --vnet-name vnet-prod \
  --name snet-app --address-prefix 10.20.2.0/24

# 3. Add the database subnet
az network vnet subnet create \
  --resource-group rg-network --vnet-name vnet-prod \
  --name snet-db --address-prefix 10.20.3.0/27

# Check what you built
az network vnet subnet list \
  --resource-group rg-network --vnet-name vnet-prod --output table
```

18. Protocols and Ports

An IP address gets data to the right machine. A port number gets it to the right application on that machine. A web server listens on port 80, an SSH server on port 22, and both can run on one machine without confusion.

TCP and UDP

	TCP	UDP
Connection	Sets one up before sending anything	Sends without setting anything up
Reliability	Confirms delivery, resends what is lost	No confirmation, no resending
Ordering	Reassembles in the correct order	Packets may arrive out of order
Speed	Slower, because of the overhead	Faster, with less overhead
Used for	Web, SSH, RDP, databases, file transfer	DNS, video streaming, voice, gaming
Analogy	A phone call - you confirm the other person is there	A postcard - you send it and hope

Ek line mein

TCP phone call jaisa hai - confirm karta hai ki pahuncha ya nahi. UDP postcard jaisa hai - bhej diya, bas. Isliye video call UDP pe chalti hai, file download TCP pe.

19. The Ports You Will Actually Open

These are the entries you will type into network security group rules for the rest of the course. Learn this table properly - it comes up in every lab and on the exam.

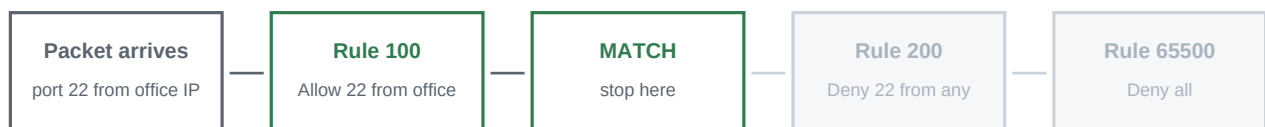
Port	Protocol	Service	Note for Azure
22	TCP	SSH	Linux administration. Opened in Lab 01.
3389	TCP	RDP	Windows remote desktop. Never leave open to the whole internet.
80	TCP	HTTP	Unencrypted web traffic.
443	TCP	HTTPS	Encrypted web traffic. What production actually uses.
53	UDP and TCP	DNS	Name resolution.
1433	TCP	SQL Server	Database tier only. Never expose to the internet.
3306	TCP	MySQL	Same warning as above.
21	TCP	FTP	Legacy and unencrypted. Avoid in new work.
389 / 636	TCP	LDAP / LDAPS	Directory services. 636 is the encrypted one.

One address worth memorising

168.63.129.16 is a special Azure address the platform uses for DNS and for health monitoring of your virtual machines. It appears in exam questions and in default rules. If you block it, things break in confusing ways.

20. How a Network Security Group Reads Your Rules

A network security group is a list of allow and deny rules attached to a subnet or to a network interface. Each rule has a priority number. Azure checks rules from the lowest number upward and stops at the first one that matches.



Lowest number is checked first. The first match wins and nothing after it runs.

Traffic on port 22 from the office address matches rule 100 and is allowed. Rule 200 never gets to run.

Priority	Name	Port	Source	Action	What happens
100	Allow-SSH-Office	22	Office IP	Allow	Checked first. Office SSH works.
200	Deny-SSH-All	22	Any	Deny	Blocks SSH from everywhere else.

Priority	Name	Port	Source	Action	What happens
65500	Default deny	Any	Any	Deny	Azure's built-in catch-all at the end.

Two consequences follow. First, a lower number wins, which reads backwards to most people. Second, if nothing matches, the default deny applies - Azure is closed by default and you open only what you need.

Ek line mein

NSG me chhota number pehle check hota hai, aur jo pehle match ho gaya wahi final hai. Aur agar kuch match nahi hua to by default block ho jaata hai.

21. Mistakes Beginners Make, and How to Avoid Them

Mistake	What happens	Do this instead
Choosing 192.168.1.0/24 for the virtual network	Collides with almost every home and office router. VPN becomes impossible.	Use a 10.x range chosen deliberately, and record it somewhere.
Forgetting the five reserved addresses	The subnet holds fewer machines than planned and has to be rebuilt.	Always add five to your machine count before choosing a prefix.
Making every subnet as small as possible	There is no room to grow and resizing needs the subnet emptied.	Private address space is free. Use /24 unless you have a reason not to.
Opening RDP or SSH to the whole internet	Automated attacks begin within hours of the VM going live.	Restrict the source to a known address, or use a bastion host.
Giving a database server a public IP	It becomes directly attackable from the internet for no benefit.	Keep it private. The application tier reaches it over the private address.
Planning subnets in the portal by trial and error	Overlaps, gaps, and a network nobody can explain later.	Plan on paper first. The portal is for typing in a finished design.

PART E

22. Drills - Binary and CIDR

Do these with a pen, not a calculator. Answers are on the next page.

Set 1 - convert to binary

a) 64 b) 100 c) 172 d) 240 e) 255

Set 2 - convert to decimal

a) 11100000 b) 10101010 c) 11111100 d) 00011000

Set 3 - how many total addresses?

a) /25 b) /27 c) /22 d) /30

Set 4 - which subnet does the address belong to?

a) 10.0.5.130/25
b) 192.168.4.70/26
c) 172.16.9.35/27
d) 10.1.1.250/28

23. Problems - Azure Scenarios

1. A subnet must hold 25 virtual machines. What is the smallest prefix that works in Azure?
2. A subnet must hold 60 virtual machines. What is the smallest prefix that works in Azure?
3. Is 172.20.5.4 public or private? Is 172.35.5.4 public or private?
4. Your virtual network is 10.50.0.0/16. Can you create a subnet 10.51.0.0/24 inside it? Why?
5. You already have subnets 10.0.1.0/24 and 10.0.2.0/24. Can you add 10.0.1.128/25?
6. Which port must be open for a Linux VM to accept an SSH connection? Which protocol?
7. An NSG has an Allow rule at priority 300 and a Deny rule at priority 150, both for port 443 from any source. What happens to traffic on 443?
8. Why should a database server usually not have a public IP address?
9. Head office uses 192.168.10.0/24. You must build an Azure network that will connect to it by VPN. Which range would you avoid, and why?
10. Split 10.0.8.0/24 into eight equal subnets. List the first three.
11. You create a /29 subnet. How many virtual machines can you actually place in it?
12. Two VMs are at 10.0.1.20/26 and 10.0.1.80/26. Are they in the same subnet?

24. Solutions

Drills

Set 1	a) 01000000	b) 01100100	c) 10101100	d) 11110000	e) 11111111
Set 2	a) 224	b) 170	c) 252	d) 24	
Set 3	a) 128	b) 32	c) 1024	d) 4	
Set 4	a) 10.0.5.128/25	block 128, blocks are 0-127 and 128-255			
	b) 192.168.4.64/26	block 64, 70 falls in 64-127			
	c) 172.16.9.32/27	block 32, 35 falls in 32-63			
	d) 10.1.1.240/28	block 16, 250 falls in 240-255			

Problems

1. $25 + 5 = 30$ addresses. /27 gives 32, which fits. Answer: **/27**.
2. $60 + 5 = 65$ addresses. /26 gives 64, not enough. /25 gives 128. Answer: **/25**. This is the trap - 60 looks like it should fit in a /26.
3. 172.20.5.4 is **private** - it falls inside 172.16 to 172.31. 172.35.5.4 is **public** - 35 is outside that range.
4. **No**. The 10.50.0.0/16 range ends at 10.50.255.255, so 10.51.0.0 is outside it entirely.
5. **No**. 10.0.1.128/25 overlaps the existing 10.0.1.0/24, and Azure rejects overlapping subnets.
6. **TCP port 22**.
7. **Denied**. Priority 150 is lower than 300, so it is checked first, and evaluation stops at the first match.
8. A public IP makes it reachable and therefore scannable from the internet. The application tier can already reach it over the private address, so the public one adds cost and risk with no benefit.

- 9.** Avoid **192.168.10.0/24** and anything overlapping it. Overlapping ranges make routing ambiguous and the VPN will not work correctly.
- 10.** Eight pieces means borrowing 3 bits, so /27 with block size 32. First three: **10.0.8.0/27**, **10.0.8.32/27**, **10.0.8.64/27**.
- 11.** A /29 has 8 addresses, Azure reserves 5, so **3 virtual machines**.
- 12. No.** Block size for /26 is 64. 20 falls in 0-63 and 80 falls in 64-127, so they are in different subnets.

25. Glossary

Term	Meaning
IP address	A 32-bit number identifying a machine on a network, written as four octets.
Octet	One of the four parts of an IPv4 address. Holds 8 bits, so 0 to 255.
Subnet mask	A 32-bit number marking which bits of an address are the network portion.
CIDR	Writing the mask as a count of network bits, for example /24.
Block size	How far apart subnet boundaries are. 256 minus the interesting mask octet.
Network address	The first address of a subnet. Identifies the subnet itself.
Broadcast address	The last address of a subnet.
Default gateway	The router a machine sends traffic to when the destination is elsewhere. In Azure, the second address of the subnet.
Private address	An address from a reserved range, not routed on the public internet.
NAT	Swapping a private source address for a public one so private machines can reach the internet.
Port	A number identifying which application on a machine should receive the data.
TCP	A protocol that establishes a connection and guarantees delivery.
UDP	A protocol that sends without establishing a connection or guaranteeing delivery.
Virtual network (VNet)	Your private, isolated network inside Azure.
Subnet	A slice of a virtual network's address space. Every resource lives in one.
NSG	Network Security Group. A prioritised list of allow and deny rules.
Peering	Connecting two Azure virtual networks. Fails if their ranges overlap.

26. One-Page Cheat Sheet

Print this page and keep it beside you during every lab.

CIDR at a glance

CIDR	Mask	Block	Total	Usable in Azure
/24	255.255.255.0	256	256	251
/25	255.255.255.128	128	128	123
/26	255.255.255.192	64	64	59
/27	255.255.255.224	32	32	27
/28	255.255.255.240	16	16	11
/29	255.255.255.248	8	8	3

Mask octet values, and private ranges

0	128	192	224	240	248	252	254	255
	/25	/26	/27	/28	/29	/30	/31	
10.0.0.0/8		10.0.0.0	to	10.255.255.255				
172.16.0.0/12		172.16.0.0	to	172.31.255.255				
192.168.0.0/16		192.168.0.0	to	192.168.255.255				

Azure reserves five addresses in every subnet

first	network identifier	second	default gateway
third and fourth	Azure DNS	last	broadcast
Sizing rule: machines needed + 5 -> pick the prefix that fits			

Common ports

22	SSH	3389	RDP	80	HTTP	443	HTTPS
53	DNS	1433	SQL	3306	MySQL	21	FTP
168.63.129.16 Azure platform DNS and health monitoring							

The three-step subnet method

1. Block size = 256 - interesting mask octet
2. Count in block-size jumps until you pass the address
3. Next block start minus one = broadcast

A virtual network is just an address range you reserved, and a subnet is a slice of it. Everything else in Azure networking is rules about what may cross those boundaries.